

INTRODUCTION TO MEASUREMENT AND DESIGN IN WEB SURVEYS

Chapters 2 and 3 examined various types of non-observation errors in Web surveys, including coverage errors, sampling errors, and nonresponse errors. The representativeness of Web samples, especially their ability to represent the general population, remains a major source of concern. However, when it comes to observation or measurement errors, Web surveys have many attractive features that may give them an advantage over other modes of data collection. At the same time, because of the broad range of design options available with Web surveys and the relative novelty of the platform, there is the risk that poor design will lead to higher levels of measurement error than with more traditional modes. In this chapter, we present an overview of Web survey measurement and discuss the role of basic design decisions in minimizing observation errors. In the next three chapters, we cover more specific design issues in greater detail. Chapter 5 examines Web surveys as a visual mode of data collection, Chapter 6 examines Web surveys as an interactive mode, and Chapter 7 examines Web surveys as a method of self-administration.

Web surveys have a lot in common with other methods of data collection. For example, mail questionnaires are also both visual and self-administered. Much of what we have learned about design through Web-based experimentation applies to other modes of data collection as well; similarly, earlier findings on the effects of the layout and design of survey questions in both interviewer- and self-administered paper surveys apply to Web surveys, too. Relevant literature on Web site design (e.g., Lynch and Horton, 2001; Nielsen, 2000) may also apply—although one should exercise caution about the wholesale adoption of design principles intended for other purposes.¹ Yet Web surveys are also distinctive. They are characterized by their great design flexibility. They can be designed in many different ways, behaving at times like computer-assisted interviews and at other times like self-administered paper

¹ For example, Nielsen (<http://www.useit.com/alertbox/application-mistakes.html>) argues for the use of default options, which make sense in e-commerce, but not in surveys.

questionnaires. Understanding the range of design elements available and using them appropriately are key issues in Web survey design.

4.1 MEASUREMENT ERROR IN WEB SURVEYS

Measurement error involves a different type of inference problem from those discussed in Chapters 2 and 3. In those chapters, we examined how well Web samples represent their target populations. In this chapter, we begin to examine how well particular observations or measurements from a Web survey respondent characterize the corresponding true values for that respondent. The simplest mathematical model for measurement error breaks each observation (y_{iA}) collected from respondent i under a specific mode (mode A) into two components—the true score and the error:

$$(4.1) \quad y_{iA} = \mu_i + \varepsilon_{iA},$$

where μ_i is the true score for respondent i and ε_{iA} is error for that respondent under that mode of data collection. To estimate the level of measurement error using this expression, we need to know the true value. Deviations from the true value can arise from a number of sources, including the respondent, the interviewer (in interviewer-administered surveys), and the instrument (including question wording, order, format, or design elements). The focus here and in Chapters 5 through 7 is on the effects of the design of Web surveys on measurement errors.

In practice, we rarely know true values. Researchers often rely on alternative approaches to examine measurement error properties of a mode or a design. One common method is to examine differences in responses to alternative presentations of the same questions, and that is the approach we rely on most often in this book. In effect, the measurement error in Equation 4.1 is broken into two components, a systematic component (M_{ijA}) due to the method of presenting the question (indexed by j) and a residual component (e_{ijA}):

$$\begin{aligned} y_{ij} &= \mu_i + M_{ijA} + e_{ijA}, \\ \varepsilon_{ijA} &= M_{ijA} + e_{ijA}. \end{aligned}$$

The classic split-ballot experiments to examine question wording effects (e.g., Schuman and Presser, 1981) are examples of this approach. One of the advantages of Web surveys is that they make it easy to carry out randomization, giving researchers a powerful tool to explore measurement effects. Still, a key drawback of this approach is that, without knowing the true values, it is sometimes difficult to say which method of presenting the question is “better”—that is, which method actually reduces error.

Another approach to examine measurement error involves administering the same (or similar) measures to respondents at different times, to explore reliability of measurement. This approach has been employed to explore mode effects (e.g., mail

versus Web), often involving repeated measurement using the same instruments (but different modes) administered to the same respondents at different times.

Whatever the approach used, researchers must often rely on indirect measures to evaluate measurement error. In Web survey design studies, these indirect measures of data quality include missing data rates, breakoff rates (potentially leading to increased nonresponse error), speed of completion, and subjective reactions by respondents. These all point to potential advantages of one design approach over another, without directly assessing measurement errors. So, in this chapter (and in the next three), we focus more on the measurement *process* than on measurement *error*, and explore how the design of Web surveys can affect both.

4.2 MEASUREMENT FEATURES OF WEB SURVEYS

Web surveys have several characteristics that affect the design of survey instruments and hence the resulting measurement error. None of these characteristics is unique to Web surveys, but each of them presents both opportunities and challenges for the Web survey designer.

First, perhaps the most striking feature of Web surveys is their rich visual character. This is a feature that has already been widely exploited by Web survey designers. Although other survey modes are visual, too—for example, pictures or images have been used in paper surveys—the ease with which visual elements can be incorporated into Web surveys is what makes the Web distinctive as a data collection mode. These visual elements include not only pictures, but colors, shapes, symbols, drawings, diagrams, and videos. The cost of adding a full-color image to a Web survey is trivial. In fact, Web surveys can allow multimedia presentation of the questions, including both sound and video. The richness of the medium brings many opportunities to enhance and extend survey measurement, and it is one of the most exciting features of Web surveys. Still, the broad array of visual and other enhancements also brings risks. Much of our research has focused on the visual aspects of Web survey design, and we explore this topic in greater detail in Chapter 5.

Second, Web surveys are computerized. Like computer-assisted personal interviewing (CAPI) and computer-assisted telephone interviewing (CATI), but unlike paper surveys, Web surveys allow a full range of advanced features to be used in questionnaires. Randomization (of question order, response order, question wording or format, etc.) is relatively easy in Web surveys. This is one reason why Web surveys have brought about a huge increase in the number of survey design experiments. Other aspects of computer-assisted interviewing that are easy to carry out in Web surveys but relatively hard to use in paper surveys include automated routing (conditional questions), edit checks, fills (inserting information from prior answers in the current question), and the like. All this means that the surveys can be highly customized to each individual respondent, based on information available from the sampling frame or from earlier questions in the survey. This permits the use of very complex instruments and complex measurement approaches, such as computerized

adaptive testing. However, adding such complexity increases the need for testing, increases the chances of programming errors, and makes careful specification and testing of the instrument all the more important.

A third feature of Web surveys, which is related to their being computerized, is that they can be designed with varying degrees of interactivity—that is, to react based on the actions of the respondent. Conditional routing is one form of interactivity. Because Web surveys can be interactive, they can be designed to behave more like interviewer-administered surveys, for example, prompting for missing data, offering clarification, providing feedback, and the like. We discuss the interactive character of Web surveys in greater detail in Chapter 6 and also discuss interactivity in the context of scrolling versus paging designs in Section 4.3 below.

A fourth characteristic of Web surveys is that they are distributed, and the designers lack control over the final display of the questions. In the traditional computer-assisted modes, the technology is in the hands of the interviewer, and both the hardware and software are provided by (and under the control of) the survey organization. This means that the designers can control the look and feel of the survey instruments. In contrast, with Web surveys, designers have little control over the browser or the hardware the respondent uses to access and complete the survey. Increasingly, people are accessing the Web using a variety of mobile devices (such as smart phones or tablets) and these present new challenges for the Web survey designer. But the presentation of Web surveys can vary in many other ways too, because of the user's setting for the size of the browser, the security settings that affect whether and how JavaScript, Flash, or other enhancements work, or the connection type that determines the speed with which respondent can download or upload information online. While hypertext markup language (HTML) is standard across browsers and platforms, JavaScript (for example) does not always behave in the same way across different operating systems. While these variations give respondents great flexibility in terms of how, when, and where they access the survey instrument, they make it difficult or impossible to ensure a consistent look and feel to the questionnaire for all respondents to the survey.

Finally, Web surveys are self-administered. In this regard, they are like paper self-administered questionnaires (e.g., mail surveys) and computerized self-administered questionnaires (e.g., computer-assisted self-interviewing [CASI] or interactive voice response [IVR]). Self-administration has long been shown to be advantageous in terms of reducing effects related to the presence of an interviewer, such as social desirability biases. At the same time, the benefits of interviewer presence—such as in motivating respondents, probing answers, or clarifying questions—are also absent. From an instrument design perspective, this means that the instrument itself must serve these functions. Each questionnaire must also be easy enough for untrained or inexperienced survey-takers to complete. Chapter 7 examines the merits of Web surveys relative to other methods of self-administration and also explores the pros and cons of fostering a greater sense of social presence in Web surveys (for example, through the use of virtual interviewers).

Given the vast range of possibilities that the Web offers, Web survey designers face many design decisions—about the general design approach (scrolling versus paging) to adopt, the response or input formats to use for specific questions, navigation, and general layout and style elements. We address each of these topics in turn below.

4.3 CHOICE OF BROAD DESIGN APPROACHES

One of the first—and most important—choices Web survey designers face is whether to present all the questions in a single scrolling Web page or to present each question on its own page. There are many intermediate options along this continuum and few surveys adopt either extreme. For example, it is common to group related questions on a page (an approach called “semantic chunking” by Norman, Friedman, Norman, and Stevenson, 2001) or to break a long scrolling survey into several pages, one for each section. Figure 4.1 presents examples of a scrolling and paging design for the same survey. Some Web survey software—especially at the low end of the market—does not support paging approaches, and other software makes it difficult to control the layout of multiple-item pages. Clearly, the decision about the design approach should not be dictated by the software, but by the objectives of the survey and the content of the questions.

Despite the importance of this basic design decision, there is remarkably little research on the relative merits of the two approaches. There are several early unpublished comparisons of the two (see, for example, Burris, Chen, Graf, Johnson, and Owens, 2001; Clark and Nyiri, 2001; Nyiri and Clark, 2003; Vehovar, Lozar Manfreda, and Batagelj, 1999), but most of these had sample sizes too small to detect meaningful differences or used instruments that would not favor either approach. Thus, it is not surprising that these studies find no differences. Since then, this topic has received little research attention. To our knowledge, only one study has explored the design choice with a large enough sample to detect differences. Peytchev, Couper, McCabe and Crawford (2006) compared a scrolling design (with the survey divided into five sections) with a paging design (with one or more questions on a page) for a survey on drug and alcohol use among college students. They found no differences in unit nonresponse, breakoffs, and nonsubstantive answers (explicit refusals or “don’t know” responses), and response distributions and key associations did not differ between versions. However, they did find significantly more item missing data and longer completion times in the scrolling version, which they attribute to the automated control of navigation and routing in the paging design.

In practice, survey researchers seem to prefer scrolling designs for short surveys with few skips or edits or for mixed-mode designs where the goal is to replicate a paper questionnaire. Overall, paging designs appear to be much more common; they give the researcher control over the order in which questions are presented, allow automated skips and routing, permit edit checks at the point of occurrence, and so on. The more complex the instrument and the more the designers want to control the interaction, the more likely a paging design will be employed.

(a) Scrolling design

Questions about this survey?
Email us at umsl@msiresearch.com
or call toll free 1.866.674.3375

Consent

About You

Tobacco & Alcohol

Other Drugs

Perceptions

Sex & Driving

These next questions have to do with TOBACCO USE

B1 IN YOUR LIFETIME, have you ever smoked a cigarette (even a puff)?

☐ Yes

☐ No [\(Skip to B11\)](#)

☐ Refused [\(Skip to B11\)](#)

B2 How old were you when you smoked your first cigarette?

☐ Refused

B3 How many cigarettes have you smoked in the PAST 30 DAYS?

☐ None [\(Skip to B11\)](#)

☐ Less than one cigarette per day

☐ 1-5 cigarettes per day

☐ About 1/2 pack per day

☐ About 1 pack per day

☐ About 1 1/2 packs per day

☐ 2 or more packs per day

☐ Refused [\(Skip to B11\)](#)

B4 Please indicate which of the following is true for you: *(Check all that apply)*

☐ I smoke mostly on weekends

☐ I smoke mostly at parties where alcohol is served

☐ I smoke mostly when I'm with friends

☐ I smoke mostly when I'm studying hard

☐ None of the above

(b) Paging design

Questions about this survey?
Email us at umsl@msiresearch.com
or call toll free 1.866.674.3375

Consent

About You

Tobacco & Alcohol

Other Drugs

Perceptions

Sex & Driving

These next questions have to do with TOBACCO USE

IN YOUR LIFETIME, have you ever...

	Yes	No	Refused
Smoked a cigarette (even a puff)	☐	☐	☐
Used smokeless tobacco (chew, snuff)	☐	☐	☐
Smoked a pipe or cigar	☐	☐	☐

Next Screen

FIGURE 4.1. Examples of scrolling and paging designs (taken from Peytchev, Couper, McCabe and Crawford, 2006). The top panel shows a scrolling design; the bottom panel a paging design. (See color insert.) Reprinted with permission from Peytchev, Couper, McCabe and Crawford, 2006.

4.4 THE LOOK AND FEEL OF THE WEB SURVEY

Another set of decisions a designer faces relates to the “look and feel” of the Web survey. With all the visual and interactive features available, there are many different design options. These decisions may be faced for an individual survey or for all surveys conducted by an organization. In either case, we believe the survey designer

should be an active participant in this process. Although they have much in common, designing a Web survey is not the same as designing a Web site. The designs serve different purposes and should encourage different types of user behavior.

Some of the elements a designer must consider include:

- Background and foreground design;
- Font or typeface;
- General layout and use of panes;
- Response requirements (forcing functions, requiring answers, etc.); and
- Navigation conventions.

Many of these design issues are discussed in greater detail in Couper (2008a, Chapter 4). We provide a brief overview here, focusing on those areas that have received the most research attention. We postpone our discussion of navigation conventions to Section 4.5.

Background/foreground design. One design decision relates to the choice of background colors, patterns, textures, or other presentation styles for each Web survey page. Paper questionnaires have typically been printed on a single color stock (usually white), although increasingly more colors are possible. For example, Dillman, Sinclair, and Clark (2003) report improvements in response rates in a 1990 census test with a design which included (among other things) a paper questionnaire printed with a light blue background with white answer spaces (see also Jenkins and Dillman, 1997). Subsequently, the 1995 National Census Test compared blue versus green census forms (again with white answer spaces), and found no significant differences in response rates (Scott and Barrett, 1996). Neither study reported any differences in data quality in or in perceived or actual completion times. The advent of Web surveys gives researchers access to potentially millions of different colors, yet the research on the effects (if any) of background color on data quality has been quite limited.

The choice of background/foreground colors has two potential implications for Web surveys. The first is its effect on legibility or readability. Insufficient contrast may make it difficult to read the survey and this effect can be exacerbated when the font size is small. While we have seen many examples of Web surveys that strain legibility, we know of no research on this topic. There are various guidelines and tools available to ensure sufficient contrast to facilitate reading online, and we can simply be prescriptive and say that these guidelines should be followed (see Couper, 2008a, pp. 164–167). Most authors favor black font on a white or light color (such as light blue or yellow) background. The second implication of the choice of background colors is the meaning that respondents may attach to the different colors (see, e.g. White, 1990) or the effect of these choices on how respondents answer the questions. Work by Schwarz and colleagues (see e.g., Novemsky, Dhar, Schwarz, and Simonson, 2007; Reber and Schwarz, 1999; Song and Schwarz, 2008a, 2008b) suggests that perceptual fluency—the ease of seeing and identifying a stimulus—can affect judgments; for example, higher levels of perceptual fluency can lead to

more positive evaluations. Perceptual fluency may be affected both by background/foreground contrast and by typeface or font size.

There are a few unpublished studies on the effect of background color in Web surveys. Pope and Baker (2005) compared white, blue, and pink backgrounds in a survey of college students. They found no significant differences in response rates or breakoff rates by background color. The actual and perceived survey times were slightly (but not significantly) lower for the blue than for the pink and white backgrounds, and these differences were larger for males than for females, suggesting that men were more negatively affected by the pink background. Baker and Couper (2007) tested white, blue, and yellow backgrounds in a consumer survey on energy use. The yellow background had a significantly higher breakoff rate (15.0 percent) than the blue background (10.8 percent) with white occupying a middle position (13.7 percent). However, background color had no significant effect on perceived or actual time of survey completion or on a variety of subjective evaluation items asked at the end of the survey. In another study, Hammen (2010) asked respondents about their preferred color for an Internet page; the largest proportion (31.1 percent) expressed no preference, but 29.5 percent preferred blue, with white being the next most popular option at 15.8 percent. She then tested white, blue, green, and red backgrounds in a Web survey and found no significant differences in breakoff rates. None of these studies reported any differences in response distributions by background color.

These three studies suggest that caution should be used in choosing background colors, but the effects are not generally large. The particular choice of a background color can be dictated by considerations of branding (i.e., matching the color scheme of the parent Web site or sponsoring organization), navigation (e.g., using different colors for different sections of the questionnaire; mapping different colors onto different actions), aesthetics, and other considerations. The evidence to date, though limited, seems to weakly favor a light neutral-colored background (e.g., light blue).

In addition to color, there are a wide variety of background visual elements that can be used in Web surveys, including textures, lines, shapes, and even images. As we show in Chapter 5, respondents tend to attribute meaning to these graphical elements. In the best case, they may distract the respondent from the task of answering the questions; in the worst, they may change the meaning of the questions. As Nielsen (2000, p. 126) argues, “Use either plain-color backgrounds or extremely subtle background patterns. Background graphic interfere with the eye’s capability to resolve the lines in the characters and recognize word shapes.”

To summarize, the background should be just that—background. The canvas on which the questions and response options are presented should be as neutral as possible, with the goal of focusing attention on the key survey tasks.

Typeface and font size. Another design issue that may affect readability of the survey questions and hence the quality of responses is the typeface and font size used. The literature on typeface is largely qualitative or descriptive (see, e.g., Lynch and Horton, 2001, Chapter 5; Schriver, 1997, Chapter 5; Spiekermann and

Ginger, 2003), and it tends to focus on reading large bodies of text rather than the relatively short questions and response options that are common in surveys. Typeface can affect the comprehension of survey questions by affecting their readability. But typeface can also convey emotional meaning, which could affect how questions are interpreted (see, e.g., Childers and Jass, 2002; McCarthy and Mothersbaugh, 2002; Novemsky et al., 2007). In general, though, as long as the font size is large enough to permit comfortable reading, and as long as the typeface is readily readable, the effect on survey response should be minimal. Further, the Web offers a potential advantage over paper instruments in that the respondents can adjust the font size setting in their browsers to facilitate comfortable reading. However, the designer must take care that permitting such flexibility does not cause other layout or design problems for any response scales, tables, or images that the survey includes.

Selective emphasis. A design decision related to the typeface involves how to emphasize certain textual elements. There are again many different possibilities, including boldface, underlining, capitalization, and color. For example, Crawford, McCabe, and Pope (2003) advocate the use of boldface for question text and regular type for the response options; they further advocate the use of blue for emphasis. There is some consensus in the Web site design literature (see, e.g., Nielsen, 2005) that color (especially blue) and underlining should *not* be used for emphasis, since these suggest linkable text in Web pages. While capitalization is not suitable for long bodies of text, it is appropriate for selective emphasis. Italicized text draws attention because of the contrast in shape from regular text, but is less readable than regular type. A common convention in Web surveys is to use italics for instructions, in part to convey the notion that they are not as important as the question text and may be ignored if not needed. If regular text is used for the questions, bold works for emphasis; otherwise, capitalization is effective. Again, the key is *selective* emphasis—too much emphasis will be counter-productive (see Lynch and Horton, 2001, pp. 132–133). Further, the use of text styles should be applied consistently throughout the instrument so that respondents can learn the meaning associated with different text elements.

Page layout and alignment. Another global decision a Web survey designer faces relates to the spatial arrangement of various elements on the screen. Because HTML is designed to facilitate a layout from left to right and from top to bottom, most designers adhere to this format, even in countries like China and Japan where vertical text may be more common in printed documents. For Western readers, and for most Internet users, the top left corner of the screen has a privileged position. The eye usually begins there when scanning Web pages (e.g., Nielsen and Pernice, 2010) and browsing activity is typically anchored to that point.

It is also common to use a header on each Web page. Headers can serve a branding or orienting function, reminding the respondent of key survey elements (topic, sponsor, etc.); they can also provide links for additional information (such as frequently asked questions, or FAQs). Research suggests that respondents quickly

Sections

Survey of State Directors of Veterans' Employment and Training Services on the Jobs for Veterans Act

Introduction

Navigation Instructions

Completing the Survey

GAO Contacts

☒ State Contact

Timeframe and Glossary

Background Information

Performance Accountability

Monitoring

Monitoring (Continued.)

Monitoring (Continued.)

Monitoring (Continued.)

Comments and Survey Completion Question

Getting a Copy of Your Responses

State Contact

Please enter the following information in the event we need to clarify a response.

Contact Name:

Title:

E-mail address:

Telephone number (area code):

< Previous section

Next section >

Print

Exit

Record 55

Questionnaire Programming Language - Version 5.0

U.S. Government Accountability Office

⏮ ⏪ ■ ⏩ ⏭ ?

FIGURE 4.2. Use of panes in a survey from the US Government Accountability Office. The pane on the left allows respondents to switch sections; the pane at the bottom allows other movements through the questionnaire. (See color insert.).

get used to the header and ignore any information it contains. This phenomenon is known as banner blindness (Benway, 1998; Benway and Lane, 1998; Pagendarm and Schaumburg, 2001). Surveys that require that navigation or other information be displayed on every page often use a vertical navigation pane. This approach appears to be more common in establishment surveys and surveys conducted by government agencies. Figure 4.2 shows an example of a survey with a pane for moving to different sections of the questionnaire (on the left) and a pane for other movements through the questionnaire (at the bottom). The careful use of background and foreground color in these panes can help keep the respondent’s attention on the key survey content.

Because different parts of the screen are more or less easily seen by respondents (see Section 5.3), it is important that the placement of the questions makes it easy for respondents to find the beginning of each question. Left-justifying the question text may help respondents locate where the question starts. With multiple questions on a page, question numbering or icons that serve to identify each question may also help guide the respondent.

There are several issues regarding the alignment and placement of the response options. One is whether the options should be arranged vertically or horizontally below the question. Some have argued that the latter approach better conveys the sense of a continuum. However, our own research (see Tourangeau, Couper, and Conrad, in press) shows no effect of the vertical or horizontal presentation of response options on response distributions, although response times are sometimes affected (see Chapter 5). Another issue is whether the input field (such as a radio button or check box) should be placed to the left or to the right of the label for the field. Arguments could be made for either position in paper surveys (see, for example, Dillman, 2007, pp. 123–124). The potential advantage for putting the input field on the right is that the skip instructions will be more visible and the labels are not obscured (for right-handed people) when the respondent answers. There is no compelling empirical evidence in favor of one over the other on the Web. In paging surveys, the skips can be handled automatically, eliminating the need for skip instructions. Further, in most Web surveys, a mouse or other indirect pointing device is used to select the answers, so that the respondent's hand does not obscure the response options (although, with the increasing adoption of touch screen devices, this may become an issue in the future). In HTML it is easier to place the input field to the left of the label, and this has become a common Web convention, so we recommend this position.

Still another alignment decision concerns the use of multiple columns. Arranging response options into several columns may reduce the need for vertical scrolling, but may lead respondents to believe that each of the different columns requires separate answers. Research by Christian, Parsons, and Dillman (2009) and Toepoel, Das, and van Soest (2009a) suggests that the arrangement of the response options into rows and columns affects response distributions. The order in which the options are read (whether by row or by column) is likely to be affected by how far apart the columns are relative to the rows. If the columns are visually distinct from each other, the respondent might read down the first column before starting on the next column. However, if the rows are more distinct, the respondent may read row-by-row. Web surveys do not have the same space restrictions as paper surveys and we see no need to use multiple columns. We agree with Christian, Parsons, and Dillman (2009, p. 420), who conclude that “presenting the options linearly in one column facilitates respondents’ processing of the scale and encourages them to process the categories in the same order, making it easier to provide a response.”

A final issue with the arrangement of response options is their relative spacing. We address this issue in more detail in Section 5.1 of the next chapter. In general, with regard to the broad design issues we have discussed here, our experience is that as long as sufficient readability is ensured and clear design conventions are followed, respondents adapt quickly to the specific design elements used in a survey. So, consistency of design may be more important than following any particular set of guidelines.

4.5 NAVIGATION CONVENTIONS

There are only a small number of actions a respondent usually performs in completing a survey, in addition to answering the survey questions. Yet, there are many different ways in which the tools used for navigation or movement through the instrument can be designed.

In scrolling surveys, the primary navigation tool is the scroll bar. This is an inherent feature of Web browsers over which survey designers have no control. The benefit of this is that most users are familiar with vertical scrolling of Web pages and require no guidance or instruction. The drawback is that the longer the Web page, the smaller the scroll bar, requiring more precise mouse movements from respondents and increasing the risk they will inadvertently miss an item.

In paging surveys, the process of moving from one page to the next is under greater designer control. Indeed, some Web survey vendors permit only forward movement, preventing users from going backwards in the instrument. Others use an automatic advance, with the next page being delivered as soon as a response option is selected (Hammen, 2010; Hays, Bode, Rothrock, Riley, Cella, and Gershon, 2010; Rivers, 2006). It is more common to have respondents press a “Next” (or “Continue” or a forward arrow) button to proceed after they have selected an answer. Requiring respondents to press the “Next” button may increase the time to answer and require extra mouse clicks, but some proportion of respondents make multiple selections in single-selection (radio button) items (see Heerwegh, 2003; Stern, 2008). We believe that using a “Next” button gives respondents the opportunity to check and reconsider their answers before proceeding. Another reason for including a “Next” button is that the automatic advance works only for single selection items, not for multiple-response questions, open-ended items, or pages with multiple questions.

So, “Next” buttons seem to be useful. What about “Previous” buttons? There is little research on this topic, but we believe giving the respondent an opportunity to go back and review or change a previous answer is likely to improve data quality. In our own surveys, we find that respondents generally do not use this function often, but when they do, it is often an indication that there was a problem understanding the earlier question that was only revealed by the follow-up question (for example, the respondent is routed to a question that does not apply to them).

If both a “Next” and a “Previous” button are included on each page of the survey, how should they be designed? There are at least three issues here: 1) whether to use the standard HTML action buttons or their graphical equivalents; 2) how to label the buttons; and 3) where to place them. We know of no research on the first topic, and our own unpublished research on the second suggests that the label (e.g., “Previous” versus “Back” versus “< –”) has little effect on performance. Couper, Baker, and Mechling (2011) experimented with the placement of the “Previous” and “Next” button (see Figure 4.3 below), and found little effect on breakoff rates or completion times, but did find effects on the level of use of the “Previous” button. Respondents were more likely to use the “Previous” button when it was to the right of and just as

prominent as the “Next” button (panels (a) and (b) in Figure 4.3). Overall, Couper and his colleagues found some support for reducing the visual prominence of the “Previous” button, either by using a hyperlink instead or placing it below the “Next” button (panels (c) or (d)). This discouraged use of the “Previous” button—which often appears to be accidental when it is prominently displayed—and reduced completion times. These findings are consistent with Wroblewski’s (2009) recommendation to place more frequently used action buttons on the left side of the screen.

4.6 CHOICE OF RESPONSE FORMATS

The design decisions we have discussed so far concern general issues that affect all pages or items in an instrument; the choice of response or input formats, by contrast,

(a) *Next* on left, *Previous* on right

Who in your household is responsible for paying your household's energy utility bills?

Select one.

- ☒ I am responsible for doing so
- ☐ Another member of my household is responsible for doing so
- ☐ Not applicable - someone who does not live in my household is responsible for doing so
- ☐ Not applicable - energy utility bills are included in rent or condominium fees the household pays
- ☐ Don't know

Next

Previous

(b) *Next* and *Previous* both on left, horizontal arrangement

Who in your household is responsible for paying your household's energy utility bills?

Select one.

- ☒ I am responsible for doing so
- ☐ Another member of my household is responsible for doing so
- ☐ Not applicable - someone who does not live in my household is responsible for doing so
- ☐ Not applicable - energy utility bills are included in rent or condominium fees the household pays
- ☐ Don't know

Next

Previous

(c) *Next* and *Previous* both on left, vertical arrangement

Who in your household is responsible for paying your household's energy utility bills?

Select one.

- ☒ I am responsible for doing so
- ☐ Another member of my household is responsible for doing so
- ☐ Not applicable - someone who does not live in my household is responsible for doing so
- ☐ Not applicable - energy utility bills are included in rent or condominium fees the household pays
- ☐ Don't know

Next

Previous

(Continued)

FIGURE 4.3. (Continued)

(d) *Next* and *Previous* both on left, hyperlink for *Previous*

Who in your household is responsible for paying your household's energy utility bills?

Select one.

- ☒ I am responsible for doing so
- ☐ Another member of my household is responsible for doing so
- ☐ Not applicable - someone who does not live in my household is responsible for doing so
- ☐ Not applicable - energy utility bills are included in rent or condominium fees the household pays
- ☐ Don't know

[Previous](#)

(e) *Previous* and *Next* both on left (*Previous* first), horizontal arrangement

Who in your household is responsible for paying your household's energy utility bills?

Select one.

- ☒ I am responsible for doing so
- ☐ Another member of my household is responsible for doing so
- ☐ Not applicable - someone who does not live in my household is responsible for doing so
- ☐ Not applicable - energy utility bills are included in rent or condominium fees the household pays
- ☐ Don't know

FIGURE 4.3. Action button conditions from Couper, Baker, and Mechling (2011) study. The different conditions vary the placement and visual prominence of the “Previous” button. Reprinted with permission from Couper, M., Baker, R., and Mechling, J. *Survey Practice*, 2011.

is specific to individual items. As with the other design decisions we have covered, there are many alternatives for capturing responses in Web surveys. In paper surveys, the response options place no absolute constraints on the respondent (who can always check multiple options or write in an answer), but the input tools in Web surveys may serve both as visual guides to the respondent about how to answer and as devices for imposing conditions on the answers (Couper, 2008b). For example, radio buttons permit the selection of one and only one response. Drop boxes or select lists similarly constrain the selection to one of the available options. This feature of input tools has the advantage of restricting the respondent to the desired action, but it places greater burden on the survey designers to ensure that all possible options are available to the respondent and that the design does not inappropriately limit the options a respondent can choose.

The set of input formats available to the survey designer may be limited to those available in HTML and those that can be created using active scripting such as JavaScript, Java, or Flash. Within HTML, there are radio buttons (commonly used for single-response questions) and check boxes (used for multiple-response or check-all-that-apply items); there are drop boxes, which allow respondents to make single or multiple selections from a long scrolling list; and there are text fields and text areas, which allow respondents to enter unformatted responses. Many Web survey

software systems have their own menu of selection tools. Using active scripting, the format of response options is almost limitless, including visual analog scales or slider bars (e.g., Couper, Tourangeau, Conrad, and Singer, 2006; Funke, Reips, and Thomas, 2011), drag-and-drop (e.g., Delavande and Rohwedder, 2008), card sort methods, map-based input, and so on. The extent to which these interactive features add value—either for the survey researcher in terms of better measurement or for the respondent in terms of greater enjoyment or smoother flow—is still a matter of debate. Software vendors whose products include these features and market research companies promoting their competitive advantage trumpet the benefits of such tools, but careful research contrasting these input tools with the more standard HTML approaches is still mostly lacking. We discuss research on one input tool—slider bars—in Chapter 6 (see Section 6.2.3).

There are two key issues for the survey designer in the selection of input formats. The first is the choice of the appropriate tool for a particular question; the second is how best to design that tool to facilitate the task of providing a response. Both decisions should exploit affordances—the notion that the shape of an object should somehow suggest how the object is to be used. Gibson (1979) first introduced the idea of affordances, arguing that visual perception involved the recognition of affordances. But Norman's (1988) classic *The Design of Everyday Things* popularized the concept and applied it to the design of interfaces. Both radio buttons and check boxes “afford” clicking, but the former permits only a single selection from a set while the latter acts as an on-off switch for each item in the set. Of course, this difference between the formats may not be apparent to inexperienced users. We continue to see examples of Web surveys where the input tool does not match the intended task. However, there is little research examining the effect of the type of selection tool on the number, quality, and completeness of responses.

In one study, Couper, Traugott, and Lamias (2001) contrasted radio buttons with text boxes for a series of questions about the race/ethnicity of the respondent's friends, classmates, etc., where the task was to enter a number from 1 to 10 for each of five groups. The text boxes allowed respondents to give invalid (out-of-range) responses, while the radio button version prevented them from giving such answers, producing significant differences in both the proportion of invalid responses and the rate of missing data (including DK/NA responses). Further, the size of the text box also affected responses, with a longer box eliciting more invalid responses than a short box. However, the text boxes made it easier for respondents to make the five numbers sum to ten (as they were instructed to). Thus, although the text entry box made it easier for respondents to avoid answering the question and failed to prevent out-of-range or otherwise unusable responses, the radio button format made it harder for respondents to make the numbers add up as requested. In part, the text boxes revealed the difficulty of the underlying task, which was not so apparent in the radio button version.

Heerwegh and Loosveldt (2002) compared radio buttons with drop boxes for a series of questions. They found no effect of format on completion rates, on nonsubstantive responses, or on missing data. They did not examine substantive distributions.

However, they did find that radio buttons required more time to download, affecting respondents with slower Internet connections. This is likely to be less of a concern given the prevalence of high-speed Internet connections today. Healey (2007) also compared radio buttons with drop boxes and reported findings similar to those of Heerwegh and Loosveldt; the input format did not significantly affect survey completions, the number of nonsubstantive answers, or overall completion times. Again, Healey did not examine substantive distributions between versions. However, drop boxes in Healey's study led to slightly higher item nonresponse and longer response times per item. Furthermore, those who used the scroll mouse to complete the survey (about 76 percent of respondents did so) were prone to change answers accidentally in the drop box condition, a phenomenon that may have been responsible for some of the inconsistencies found in the 2009 Canadian Census test (Lebrasseur, Morin, Rodrigue, and Taylor, 2010).

Couper, Tourangeau, Conrad, and Crawford (2004a) examined response order effects using three different formats: a series of radio buttons, a drop box, and a scroll box (with a partial list of the response options visible). They found that the magnitude of the response order effects depended on the format used to display the items, with significantly bigger order effects in the scroll box version. Their findings suggest that while reading order produces primacy effects in visually presented measures, the effect is exacerbated when a respondent action is required to see the last few options in the list. Respondents were much more likely to select one of the options visible from the outset than to scroll to view and select one of the other response options. We discuss this study further in Section 5.3 in the next chapter.

Aside from these studies, there are few empirical tests of alternative response formats. Much more research has focused on the *design* of such input tools; we review this work in Chapter 5.

4.7 GRID OR MATRIX QUESTIONS

Whether the survey uses a scrolling or paging design, a design issue for many surveys is whether to group items that share a common set of response options into a grid or matrix. The use of grid questions is common practice in Web surveys, whether in market research, academic, or government settings. Yet, the research on grid questions suggests that, although grids may reduce completion time, the format may also increase breakoffs, missing data, and measurement error. Whether this means that grids are inherently problematic or that they are often poorly designed is the focus of current research.

Research on grids can be grouped into two categories, the first comparing questions in grids to questions in other formats and the second exploring the effects of alternative grid designs on respondent performance on the grid questions. We first examine the research on grids versus other methods for presenting the items. A summary of these studies is presented in Table 4.1.

TABLE 4.1. Studies on Grids versus Alternative Designs

Study	Sample and design	Key Findings
Couper, Traugott, & Lamias (2001)	U.S. college students; five items in one grid versus five separate pages; 11 items in three grids (four, four, and three per page) versus 11 separate pages; n = 665	• Grids significantly faster ($p < .05$) • Significantly less missing data in grids ($p < .01$) • Inter-item correlations not significantly different
Bell, Mangione, & Kahn (2001)	General population volunteers recruited on Web sites in the U.S.; Grid versus expanded format (item by item on same page) for each scale of the SF-36; n = 1,464	• Response times not significantly different • Inter-item correlations not significantly different
Tourangeau, Couper, & Conrad (2004)	Members of U.S. opt-in panel; eight agree-disagree items on seven-point scale; items in one grid versus two grids (four items each) versus eight separate pages; n = 2,568	• Significant linear trend of inter-item correlations ($p < .01$): One grid > two grids > separate items • Significant ($p < .01$) trend in differentiation: One grid < two grids < separate items • Significant ($p < .001$) trend in completion time: One grid < two grids < separate items
Yan (2005)	Members of U.S. opt-in panel; six items in one grid versus six items on same page in item-by-item format versus six separate pages; also varied instructions on relatedness of items; n = 2,587	• Inter-item correlations not significantly different • Perceived relatedness not significantly different
Toepoel, Das, & van Soest (2009b)	Probability panel in the Netherlands; 40 agree-disagree items on five-point scale; items in one grid versus four grids (ten items in each) versus ten grids (items in each) versus 40 separate pages; n = 2,565	• No effect of format on mean score or variance • Inter-item correlations not significantly different • Significant ($p < .01$) increase in missing data as number of items on page increases • Significant ($p = .006$) decrease in time as number of items on page increases • Significantly ($p < .001$) lower evaluations as number of items on page increases

(Continued)

TABLE 4.1. (Continued)

Study	Sample and design	Key Findings
Callegaro, Shand-Lubbers, & Dennis (2009)	Probability-based panel in the U.S.; ten items on five-point scale; grid versus single item per screen (collapsing across other conditions); n = 1,419	• Faster completion time for grid (median = 45 seconds) than single items (median = 70 seconds) • Inter-item correlations not significantly different • Correlations between reverse-worded items and rest of scale not significantly different • No significant differences in subjective enjoyment

Couper, Traugott, and Lamias (2001) examined a five-item knowledge measure (with the items on one page in a grid or on five separate pages) and an 11-item attitude measure (with the items in three grids or on 11 separate pages). The grid version took significantly less time to complete (168 seconds on average for the 16 items) than the item-by-item version (194 seconds). The inter-item correlations (as measured by Cronbach’s alpha) were slightly but not significantly higher for the grid versions of the scales. The researchers also found significantly lower rates of item missing data (DK or NA responses) in the grid version.

In another early experiment, Bell, Mangione, and Kahn (2001) contrasted a grid version of the SF-36 health questionnaire with an item-by-item version (with all items on the same page). Completion time for the item-by-item version was slightly longer than for the grid version (5.22 versus 5.07 minutes), but the difference was not statistically significant. They found no differences in inter-item correlations (again using Cronbach’s alpha).

Tourangeau, Couper, and Conrad (2004) compared eight agree-disagree items presented in three formats: 1) all the items on a single page in one grid, 2) four items in each of two grids presented on separate pages, and 3) each item on a separate page. Respondents took significantly less time to answer in the grid format (60 seconds on average) than when each item was on a separate page (99 seconds). They also found a significant linear trend in the alpha coefficients across the three conditions, with inter-item correlations increasing as the grouping of items increased. However, they found that respondents who got the items in a grid showed less differentiation in the answers—that is, they were more likely to choose the same response option for all items. Further, the part-whole correlations for two reverse-worded items were weaker in the grid version, suggesting that respondents were less likely to notice the reverse wording when the items were in a grid. A reanalysis of these data by Peytchev (2005) using structural equation modeling suggests that the increased correlations indicated higher measurement error rather than increased reliability of measurement. These results suggest that the faster completion times may come at the cost of suboptimal responding.

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Yan (2005) explicitly tested whether putting items in a grid encouraged respondents to see them as more highly related. She assigned respondents to one of three versions of a set of six loosely-related items: 1) one item per page, 2) all six items on the same page in an item-by-item approach, and 3) the six items in a single grid. She also manipulated the introduction to the six items, with one introduction suggesting all the items came from the same source and another suggesting they came from different sources. The effect of the layout on the inter-item correlations (Cronbach's alpha) was not statistically significant. In a follow-up question asking respondents how related they thought the items were, she again found no significant effect of layout.

Toepoel, Das, and van Soest (2009b) compared four versions of a 40-item arousal scale, with one, four, ten, or 40 items per page. They found no effect of format on the arousal index scores and only modest effects on inter-item correlations. However, item missing data increased monotonically with the number of items on a page. Using grids reduced the time to complete the set of items, but also lowered scores on respondents' subjective evaluations of the questionnaire.

Finally, Callegaro, Shand-Lubbers, and Dennis (2009) tested two subscales of the SF-36 health questionnaire in grid versus single-item-per-page versions, also varying the shading of rows. They report median response times of 45 seconds for the grid groups and about 70 seconds for the single item per screen groups. The inter-item correlations (Cronbach's alpha) did not differ significantly by version nor did the reverse-worded items show weaker correlations in the grid conditions. Further, they found no significant differences in the perceived difficulty or self-reported enjoyment of respondents across the versions.

Perhaps the clearest finding from these studies is that grids reduce the time it takes for respondents to complete a set of items. Four of the six studies find significant reductions in response times with grids. Tourangeau and his colleagues also find significantly higher inter-item correlations with grids, but none of the other studies replicates this finding. The studies also disagree about the impact of grids on item nonresponse. One study reports significantly lower item nonresponse when the items are in a grid format (Couper, Traugott, and Lamias, 2001), but another finds higher item nonresponse with longer grids (Toepoel, Das, and van Soest, 2009b). These inconsistencies across studies may be explained by a number of factors, such as the topics or populations in the study, whether the items asked about behaviors or attitudes, whether any of the items were reverse-worded, the number of items (rows) in the grid, the number of response options (columns), and so on.

Grids remain popular despite research suggesting they may be problematic for respondents, and researchers are exploring how the design of grids can be improved to mitigate their negative effects while still taking advantage of their virtues; we discuss this work in Section 6.2.5. Taken together, these studies suggest that the complexity of grids may be an important factor contributing to their negative effects on survey breakoffs, item missing data, and respondent satisfaction. If grids are to be used, making them simpler—whether by reducing the number of items (rows) in the grid, by splitting the questions (reducing the number of columns), or by using visual

feedback to guide respondents as they answer grid questions—may reduce some of their negative effects.

4.8 SUMMARY

There is a long history of research on the wording of survey questions. More recently, attention has turned to other aspects of questionnaire design, particularly in Web surveys where there are so many design options available and where the respondents interact directly with the instrument rather than through a trained interviewer. Given the vast number of design options, it is not surprising that there is still much work to be done to understand when, how, and why the different options may affect the answers obtained in Web surveys. Much of the research done so far has examined differences in item missing data rates, completion times, or response distributions. Assumptions need to be made about which designs yield more accurate data or improve validity. Further, relatively little has been done to explore the reliability of measurement under different design conditions.

Many of the experiments described in this chapter—and in the next three chapters—are based on studies of college students or of members of non-probability panels. The former are likely to be quite adept at using the Internet and to respond well to unexpected designs. The latter generally get large numbers of survey requests, often from different vendors using different layouts and designs. These veteran respondents may have become inured to design variations because of their exposure to a wide array of good and bad Web survey designs. It is, therefore, important to replicate findings about Web survey design with members of probability panels. We have done this with the Knowledge Networks and FFRISP panels; similarly, Toepoel and her colleagues (2009a, 2009b) have done this using the CentERpanel. These replications give us greater confidence that the results we summarize here generalize across other groups with different Web survey experiences or different motivations (see also Toepoel, Das, and van Soest, 2008, 2009c). But we should remain wary about the possible influence of the samples in our studies on the effects—as well as the lack of effects—that we find. It is impossible to completely separate concerns about measurement error from concerns about representativeness and population inference.

Regardless of the shortcomings in the evidence, it seems clear that careful attention to the design details may improve the quality of the answers obtained and may reduce breakoffs and missing data in Web surveys. There is no simple recipe for achieving an effective Web survey design, and researchers still have great flexibility in designing Web surveys so long as they stay within some broad limits. Still, the evidence presented in this chapter suggests that the overall design of the instrument should receive as much attention as the content.